

ELECTRICAL & ELECTRONICS TERMS

Quick Reference Tables for First-Year Engineering Students

How to use this document: Each term is explained in one simple line, with its common use, unit, and symbol/notation. “—” means there is no single SI unit or universal symbol. Circuit-diagram symbols can vary slightly by standard.

1. Basic Electrical Quantities

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Electric charge	A property of matter that causes electrical effects.	Describes the amount of electricity transferred or stored.	coulomb (C)	Q
Electric current	The rate at which electric charge flows.	Measures flow of charge in a circuit.	ampere (A)	I
Voltage / potential difference	The energy transferred per unit charge between two points.	Drives current through a circuit.	volt (V)	V
Electromotive force (EMF)	The energy supplied per unit charge by a source.	Describes the voltage provided by a battery or generator.	volt (V)	E or ϵ
Resistance	The opposition a material offers to current flow.	Limits current and sets voltage drops.	ohm (Ω)	R
Conductance	A measure of how easily current flows.	Expresses the inverse of resistance.	siemens (S)	G
Resistivity	A material property describing how strongly it resists current.	Helps compare materials for conductors and insulators.	ohm-metre ($\Omega\cdot\text{m}$)	ρ
Conductivity	A material property describing how easily it conducts current.	Helps compare conducting materials.	siemens per metre (S/m)	σ
Electrical power	The rate at which electrical energy is transferred or converted.	Rates the power used or delivered by a device.	watt (W)	P
Electrical energy	The total electrical work transferred over time.	Measures energy consumption or delivery.	joule (J); commercial use: kWh	E or W
Frequency	The number of cycles completed per second.	Describes how quickly an AC signal repeats.	hertz (Hz)	f
Time period	The time taken to complete one cycle.	Describes the duration of one waveform cycle.	second (s)	T
Phase angle	The angular position of a waveform relative to a reference.	Compares timing between AC signals.	radian (rad) or degree ($^\circ$)	ϕ or θ
Efficiency	The ratio of useful output to total input.	Compares how effectively a device converts energy.	dimensionless; often %	η

2. Circuit Elements and Sources

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Resistor	A component designed to provide resistance.	Limits current or divides voltage.	ohm (Ω)	R
Capacitor	A component that stores energy in an electric field.	Used for filtering, coupling, timing, and energy storage.	farad (F)	C
Inductor	A component that stores energy in a magnetic field.	Used in filters, converters, and magnetic circuits.	henry (H)	L
Voltage source	A source that maintains a specified voltage across its terminals.	Supplies electrical potential to a circuit.	volt (V)	Vs
Current source	A source that supplies a specified current.	Provides a set current to a circuit.	ampere (A)	Is
Independent source	A source whose value does not depend on another circuit variable.	Models a specified supply voltage or current.	V or A, as applicable	—
Dependent source	A source whose value depends on another voltage or current in the circuit.	Models controlled sources and active circuits.	V or A, as applicable	—
Active element	An element that can supply energy to a circuit.	Examples include sources and powered electronic devices.	Depends on quantity	—
Passive element	An element that absorbs or stores energy but does not generate it.	Examples include resistors, capacitors, and inductors.	Depends on quantity	—

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Open circuit	A circuit path that is broken, so current cannot flow through that path.	Represents a switch-off or disconnected path.	—	—
Short circuit	A connection with very low resistance between two points.	May bypass a component; unintended shorts can be hazardous.	ohm (Ω), ideally 0 Ω	—
Ground / reference node	The node chosen as the zero-voltage reference.	Provides a common voltage reference for circuit analysis.	volt reference (0 V)	GND / $\varnothing$

3. Circuit Laws and Network Quantities

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Ohm's law	For an ohmic conductor at constant physical conditions, voltage equals current multiplied by resistance.	Relates voltage, current, and resistance.	—	$V = IR$
Kirchhoff's Current Law (KCL)	The total current entering a node equals the total current leaving it.	Analyzes currents at circuit junctions.	—	$\Sigma I = 0$
Kirchhoff's Voltage Law (KVL)	The algebraic sum of voltages around a closed loop is zero.	Analyzes voltages around circuit loops.	—	$\Sigma V = 0$
Series circuit	Components are connected in one path, so the same current flows through each.	Used where a single current path is required.	—	—
Parallel circuit	Components are connected across the same two nodes and have the same voltage.	Provides multiple current paths.	—	—
Equivalent resistance	A single resistance that has the same terminal behaviour as a resistor network.	Simplifies circuit calculations.	ohm (Ω)	R_{eq}
Voltage divider	A series network that provides a fraction of its input voltage at an output point.	Produces a lower voltage from a higher one.	volt (V)	—
Current divider	A parallel network in which source current divides among branches.	Determines branch currents in parallel circuits.	ampere (A)	—
Impedance	The total opposition to AC current, including resistance and reactance.	Analyzes AC circuits.	ohm (Ω)	Z
Reactance	The opposition to AC caused by capacitance or inductance.	Describes capacitor and inductor effects in AC circuits.	ohm (Ω)	X
Admittance	The measure of how easily an AC circuit allows current to flow.	Used in AC network calculations.	siemens (S)	Y
Power factor	The ratio of real power to apparent power in an AC circuit.	Indicates how effectively apparent power is converted to useful work.	dimensionless	PF or $\cos \varphi$

4. AC and Waveform Terms

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Alternating current (AC)	Current that changes magnitude and reverses direction periodically.	Used in mains power and AC systems.	ampere (A)	$i(t)$ or I
Direct current (DC)	Current that flows in one direction.	Used in batteries and most electronic circuits.	ampere (A)	I
Peak value	The maximum instantaneous magnitude reached by a waveform.	Specifies the maximum signal level.	V, A, etc., depending on quantity	V_p or I_p
Peak-to-peak value	The difference between the positive and negative peaks.	Describes the full waveform excursion.	V, A, etc.	V_{pp} or I_{pp}
RMS value	The effective AC value that produces the same heating effect as an equivalent DC value in a resistor.	Compares AC voltage or current with DC ratings.	V or A	V_{rms} or I_{rms}
Average value	The mean value of a waveform over a stated interval.	Summarizes a signal's average level.	V or A	V_{avg} or I_{avg}
Amplitude	The magnitude of a signal measured from its reference level.	Describes signal strength.	V, A, etc.	A or amplitude notation
Wavelength	The distance a wave travels during one	Describes spatial repetition in wave	metre (m)	λ

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
	complete cycle.	propagation.		
Duty cycle	The percentage of one period for which a periodic signal is active.	Controls average output in switching and pulse circuits.	dimensionless; often %	D
Phase	The position of a periodic waveform within its cycle.	Compares timing between periodic signals.	radian (rad) or degree (°)	φ

5. Semiconductor and Electronic Components

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Semiconductor	A material whose electrical conductivity lies between that of a conductor and an insulator.	Forms the basis of diodes, transistors, and integrated circuits.	—	—
PN junction	The boundary formed where p-type and n-type semiconductor regions meet.	Forms the operating junction in a diode.	—	—
Diode	A semiconductor component that mainly allows current to flow in one direction.	Used for rectification, protection, and signal detection.	current in A; voltage in V	D; circuit symbol varies
Zener diode	A diode designed to operate in reverse breakdown within its rating.	Used for voltage reference and regulation.	volt (V)	ZD
Light-emitting diode (LED)	A diode that emits light when forward current flows.	Used for indicators, displays, and lighting.	current in A; forward voltage in V	LED
Photodiode	A semiconductor diode that converts incident light into an electrical signal.	Used in light sensing and optical communication.	ampere (A) for photocurrent	PD
Transistor	A semiconductor device used to control current or amplify signals.	Used as an amplifier or electronic switch.	—; terminal voltages/currents use V/A	Q or T
BJT	A bipolar junction transistor controlled by base current and terminal voltages.	Used for switching and amplification.	—	Q; terminals B, C, E
MOSFET	A field-effect transistor controlled primarily by gate-to-source voltage.	Used for electronic switching and amplification.	—	Q; terminals G, D, S
Integrated circuit (IC)	A complete electronic circuit fabricated on a small semiconductor chip.	Performs functions such as processing, amplification, and timing.	—; ratings vary	IC / U
Operational amplifier (op-amp)	A high-gain differential amplifier used with external feedback components.	Used for amplification, filtering, and signal conditioning.	volt (V) output; input/output ratings vary	Op-amp triangle symbol
Transformer	A device that transfers AC energy between windings by electromagnetic induction.	Changes AC voltage/current levels and provides isolation.	volt (V), ampere (A), VA	T
Relay	An electrically operated switch.	Allows one circuit to control another.	coil voltage (V), contact current (A)	K or relay symbol
Switch	A component that opens or closes an electrical path.	Turns a circuit or branch on and off.	current (A), voltage (V) ratings	S or SW
Fuse	A protective device that opens a circuit when excessive current melts its element.	Protects wiring and equipment from overcurrent.	ampere (A) rating	F or fuse symbol

6. Instruments and Measurement

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Ammeter	An instrument used to measure electric current.	Connected in series with the branch being measured.	ampere (A)	A
Voltmeter	An instrument used to measure potential difference.	Connected in parallel across the two points.	volt (V)	V
Ohmmeter	An instrument used to measure resistance.	Checks resistance when the circuit is safely de-energized.	ohm (Ω)	Ω
Multimeter	An instrument that combines several electrical measurement functions.	Commonly measures voltage, current, and resistance.	Depends on selected measurement	DMM
Oscilloscope	An instrument that displays voltage variation	Observes waveform shape, amplitude, period,	volts (V) and	CRO / DSO

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
	with time.	and timing.	seconds (s) scales	
Wattmeter	An instrument used to measure electrical power.	Measures power consumed or delivered by a circuit.	watt (W)	W
Energy meter	An instrument that measures electrical energy consumption.	Used to measure energy used by a building or device.	kilowatt-hour (kWh)	kWh meter

7. Digital Electronics Terms

Term	Simple one-line meaning	Use / application	Unit	Symbol / notation
Binary number system	A number system that uses only the digits 0 and 1.	Represents data in digital circuits.	—	Base 2
Bit	The smallest unit of digital information, with a value of 0 or 1.	Represents one binary state.	—	b
Byte	A group of eight bits.	Commonly represents a unit of digital data storage.	byte (B)	B
Logic 0 / LOW	A logic state interpreted as the lower voltage range for a digital input.	Represents the low state in digital logic.	voltages depend on logic family	0 / LOW
Logic 1 / HIGH	A logic state interpreted as the higher voltage range for a digital input.	Represents the high state in digital logic.	voltages depend on logic family	1 / HIGH
Logic gate	A digital circuit that performs a Boolean operation on input signals.	Builds decision-making and processing circuits.	—	AND, OR, NOT, etc.
AND gate	Produces HIGH only when all inputs are HIGH.	Implements logical conjunction.	—	AND
OR gate	Produces HIGH when at least one input is HIGH.	Implements logical disjunction.	—	OR
NOT gate	Produces the opposite logic state of its input.	Inverts a digital signal.	—	NOT / inverter
NAND gate	Produces the inverse of the AND operation.	Used as a universal logic gate.	—	NAND
NOR gate	Produces the inverse of the OR operation.	Used as a universal logic gate.	—	NOR
XOR gate	Produces HIGH when its two inputs are different.	Used in adders and parity circuits.	—	XOR
Flip-flop	A bistable digital circuit that stores one bit.	Stores state in registers and sequential circuits.	—	FF
Microprocessor	A programmable processor implemented on an integrated circuit.	Executes instructions to process data and control operations.	clock frequency in Hz	MPU / μ P
Clock signal	A periodic signal used to coordinate digital operations.	Synchronizes sequential digital circuits.	hertz (Hz)	CLK